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Session : Presentation of Select Technical Papers

Strengthening Underground Cable N/w for Reliable and Flexible distribution system

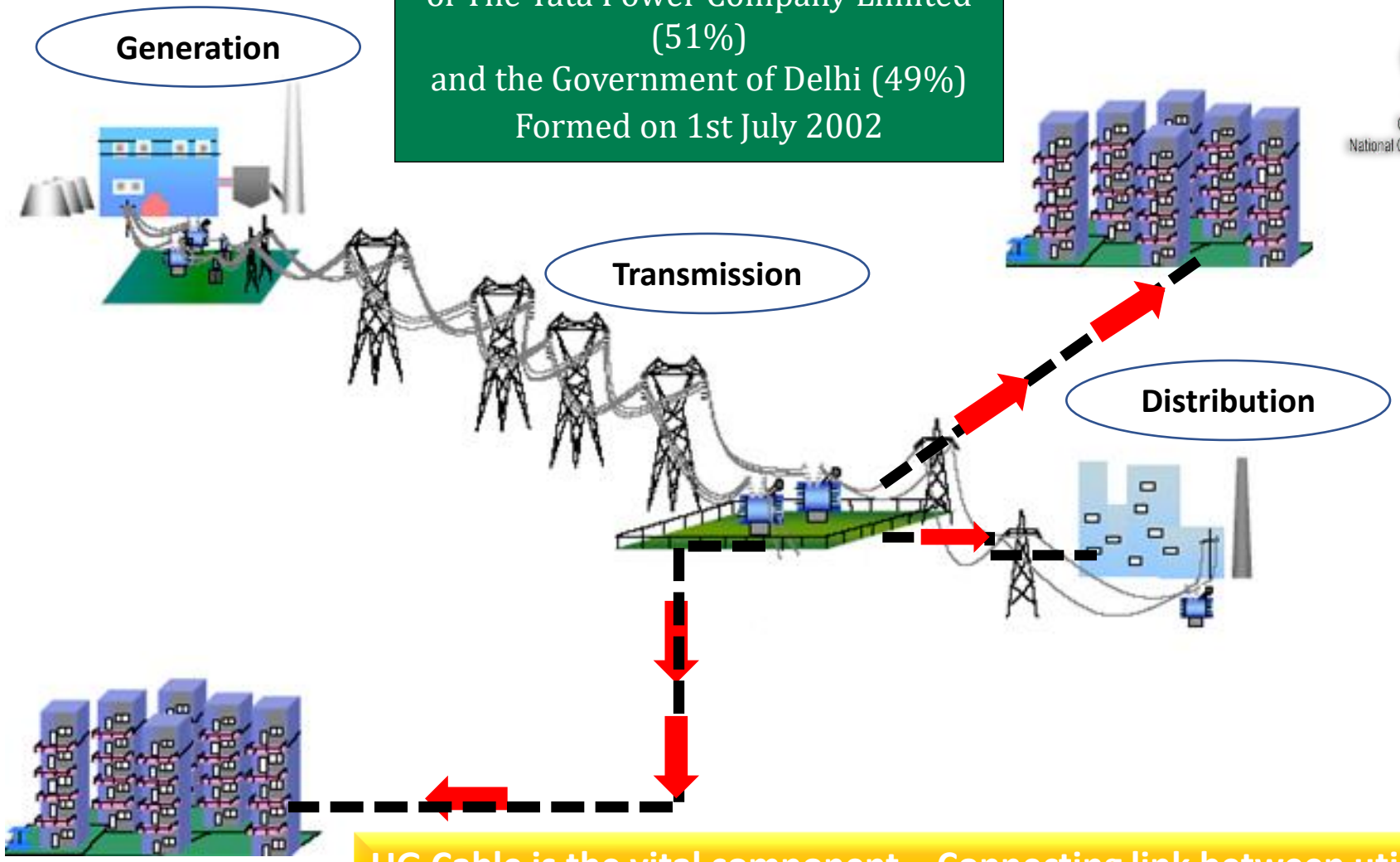
Presented By

Vineet Tripathi, AGM & Ashutosh Jaiswal, Asst. Manager, Tata Power DDL

Introduction

Significance of Power cables

A Joint Venture
of The Tata Power Company Limited (51%)
and the Government of Delhi (49%)
Formed on 1st July 2002



TATA Power-DDL

- 510 Sq. Km area
- 2.1 Million Customer Base
- 3100ckt. Km HV/EHV cable

UG Cable is the vital component – Connecting link between utility & Consumer

Underground Cable laying Challenges

UG Cable laying in TPDDL annually

- 150ckt KM 11kV
- 40ckt KM 33kV

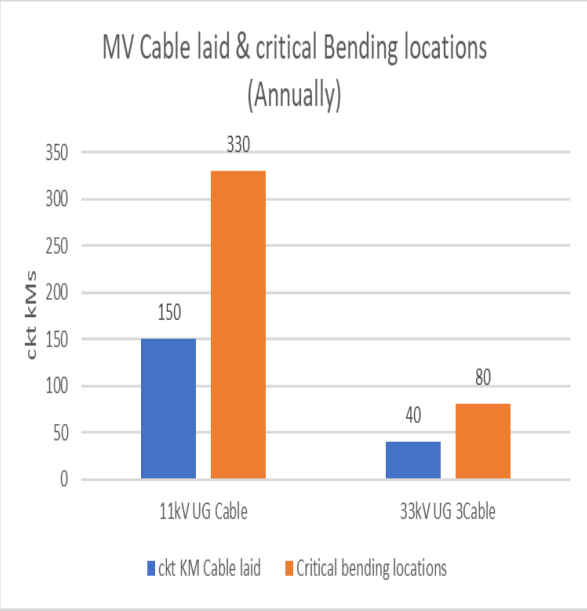


TABLE I. BENDING RADIUS OF PVC & XLPE

Voltage Grade	Min Bending Radius
3.3 Up to 11 kV Cables	15xD for Single & multicore
Above 11 kV	15xD for multicore Cables & 20xD for Single Cable

*Where D is overall Diameter of the

5.3 Minimum Permissible Bending Radii The cable should not be bent to a sharp radius. Minimum recommended bending radii are given in Table 5 (*see* Fig 1). Wherever possible, larger bending radii should be used

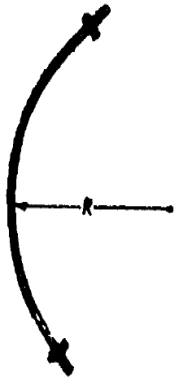
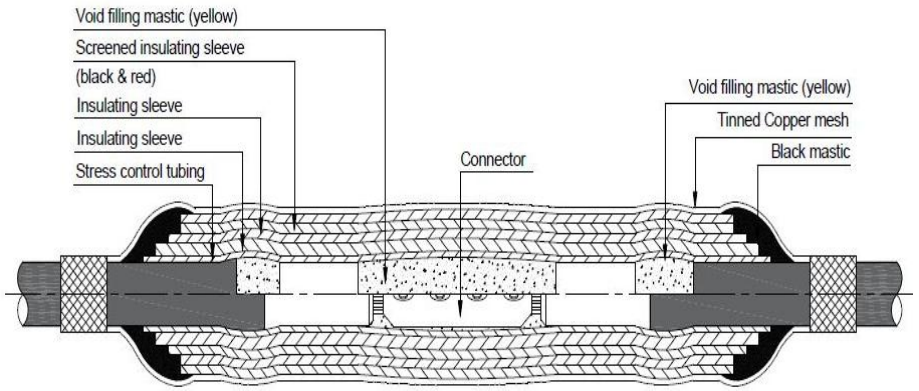
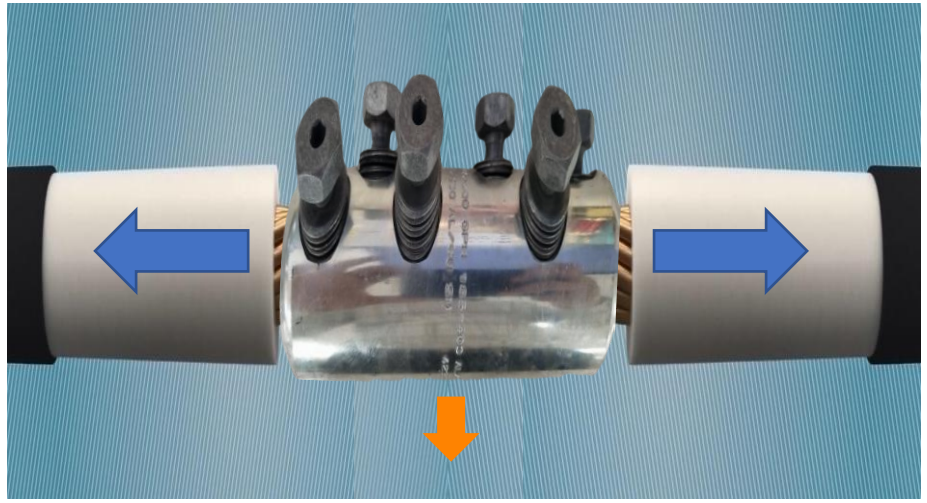


FIG. 1 PERMISSIBLE BENDING RADII

R DELHI DISTRIBUTION LIMITED - NEW DELHI ON 07/1



MV Cable Joint

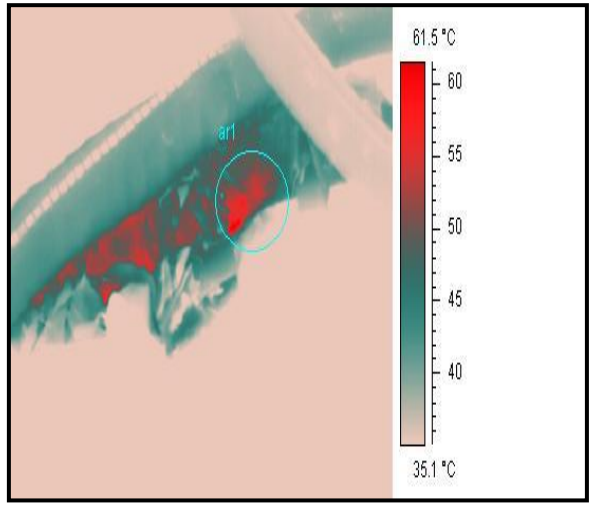


Pulling Force exerted on joint due to support removal from beneath cable

Relevance

Installation Challenges:

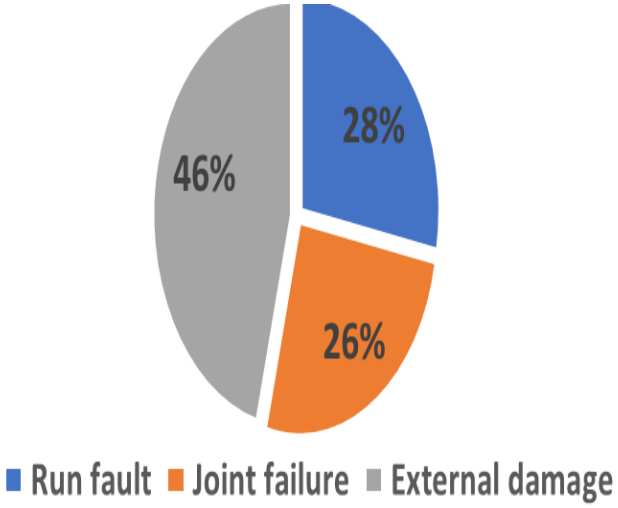
Site image of Cable Bend Location

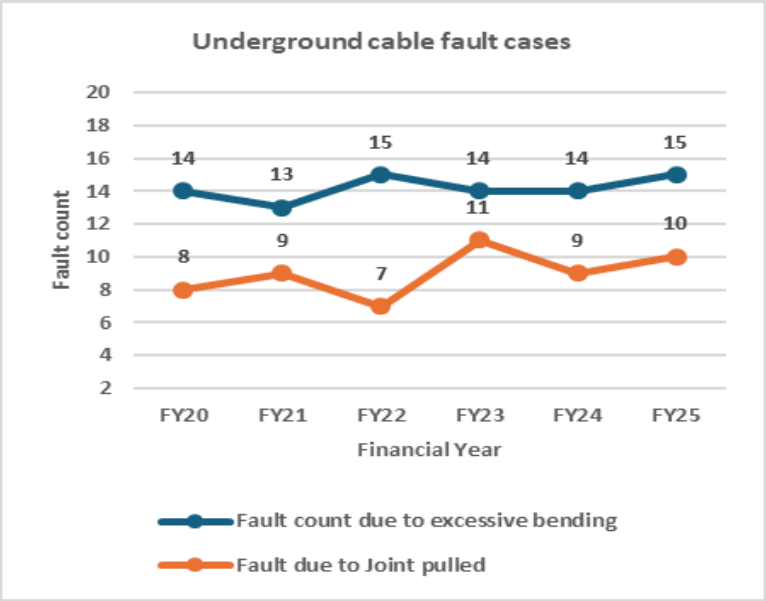


External Damages and Environmental Challenges:

Site image of Cable Joint pulled

UG Cable n/w fault





Fault Analysis

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Fault development due to excessive cable bending

- Excessive bending often violating IS 7098 MBR guidelines (15×D) during installation
- *Elongation: polymer chains are pulled apart*
- *Micro-voids form at crystalline–amorphous boundaries in XLPE*
- *These voids increase the local electric field (E-field)*
- *Local field enhancement initiates partial discharges (PD)*
- Electrical treeing and breakdown to fault

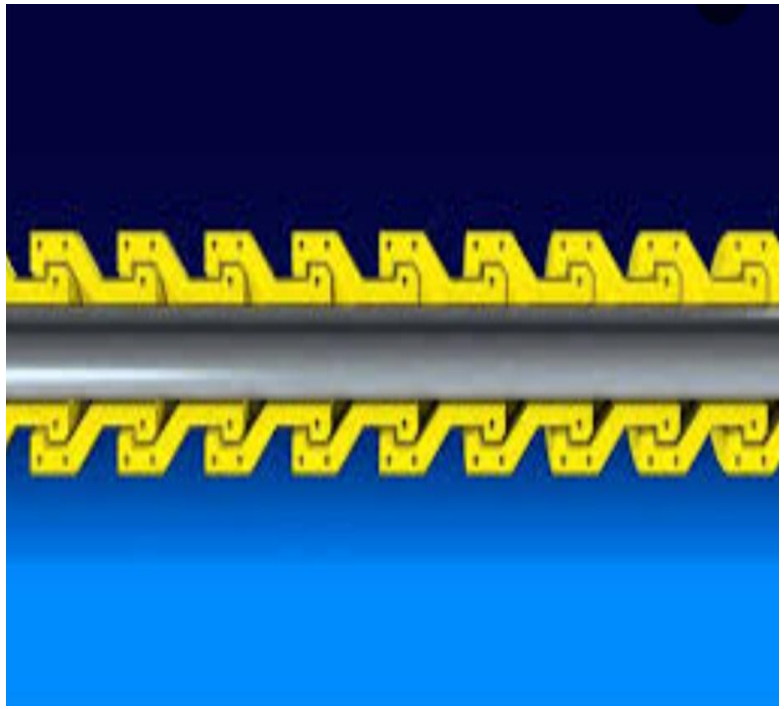
Fault development due to Cable joint pull reason

- Cable Joint Pulling due to Soil removal
- *Stress on cable connector and adhesive tubes*
- *Loosening of Conductor contact and Air gap in tubes*
- *Increased resistance of joint & Chances of Water Ingress*
- *Joint heating and oxidation due to moisture*
- *Conversion to Fault*

UG Cables Bend restriction and Cable Joint protection

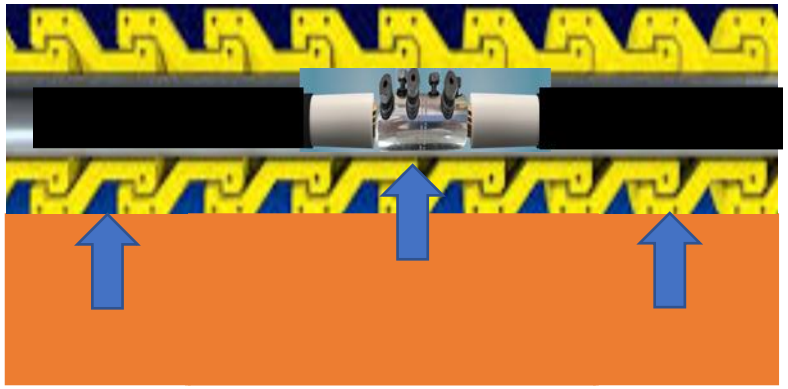
We have identified an unique solution for Underground Cables and Cable Joint protection

Usage of Cable bend restrictor at critical bending locations & Joint protection



Interlocking mechanism of "Cable Bend Restrictor"

- Provides mechanical protection to cable.
- Designed to Limit the bending radius within permissible limit as per IS 1255 requirement
- Provides support to straight through joint in Underground cable.
- Lightweight solution easy to install.



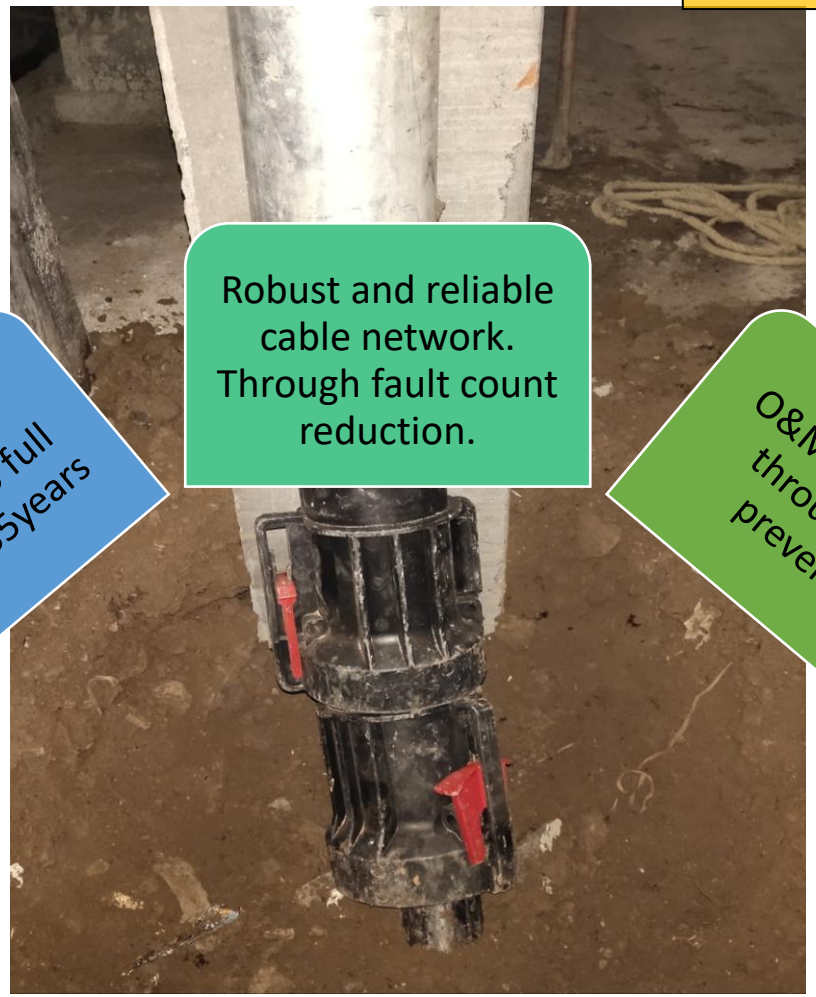
Use Case

Before

Insures cable lifespan up to full it's life i.e. 35years

Sharp cable bending at Pole

After



Robust and reliable cable network. Through fault count reduction.

O&M Cost saving through fault prevention

Usage of Cable bend restrictor at cable raising on pole
(During and after installation pic)

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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

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